K- Means using centroid (Two Dimensional objects)

Object	Attribute 1(X)	Attribute 2(Y)
A	2	2
B	3	2
C	1	1
D	3	1
E	1.5	0.5

Using Euclidean Distance

	A	B	C	D	E
A	**0**	**1**	**1.41**	**1.41**	**1.58**
B	**1**	**0**	**2.24**	**1**	**2.12**
C	**1.41**	**2.24**	**0**	**2**	**0.71**
D	**1.41**	**1**	**2**	**0**	**1.58**
E	**1.58**	**2.12**	**0.71**	**1.58**	**0**

1. Initial value of centroids A and C.
   Let C1 and C2 denote the coordinate of the centroids.
   - C1 (2,2)
   - C2(1,1)

2. Objects - Centroids distance: Calculate the distance between cluster centroid to each object

$D^0 =$

A	B	C	D	E	Cluster centroid
0	1	1.41	1.41	1.58	C1(2,2)(A) - G1
1.41	2.24	0	2	0.71	C2(1,1)(C) - G2

3. Objects Clustering : Assign each object based on the minimum distance

$G^0 =$

A	B	C	D	E	Cluster centroid
1	1	0	1	0	C1(2,2)(A) - G1
0	0	1	0	1	C2(1,1)(C) - G2

4. Iteration 1 , determine centroids

Group1 has 3 members .

Group2 has 2 members .

C1 = ((2+ 3 + 3)/3 , (2+2+1) /3) = (2.67 , 1.67)

C2 = ((1+1.5) / 2 , (1+ 0.5) /2)) = (1.25 ,0.75)

5. Itération – 1, Object Centroids distance

D^1 =

A	B	C	D	E	Cluster Centroid
0.75	**0.47**	**1.79**	**0.75**	**1.65**	**C1(2.67,1.67) - G1**
1.41	**2.24**	**0.32**	**1.76**	**0.35**	**C2(1.25,0.75) - G2**

6. Iteration , Objects clustering

$G^1 =$

A	B	C	D	E	Cluster centroid
1	1	0	1	0	C1(2.67,1.67) - G1
0	0	1	0	1	C2(1.25,0.75) - G2

Comparing the groups of G1 and G2 there is no change in the group members.

So final clusters are
Group 1 = { A, B & D }
Group2 = { C, E}